

Education by a dietitian in patients with heart failure results in improved adherence with a sodium-restricted diet: A randomized trial

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Background Multidisciplinary heart failure (HF) programs reduce hospital readmission and improve clinical outcomes. Although dietitians are often members of such teams, no randomized studies have demonstrated the independent benefit of dietitian-administered dietary counseling for patients with HF. The purpose of this study was to evaluate the effect of dietitian education on adherence to a sodium-restricted diet in ambulatory patients with stable HF.

Methods Patients with HF (left ventricular ejection fraction <35%) were randomized into a dietitian education group (n = 23) or a usual care group (n = 24), then observed for 3 months. Both groups received a 2 g/d dietary sodium prescription. The usual care group received nutrition advice by way of self-help literature, whereas the dietitian education group returned for 2 counseling sessions with a dietitian.

Results Dietitian education resulted in a significant decrease in sodium intake at 3 months (2.80 ± 0.30 to 2.14 ± 0.23 g/d, $P < .05$). In contrast, there was no change in sodium intake in the usual care group (3.00 ± 0.31 to 2.74 ± 0.35 g/d, $P = \text{ns}$).

Conclusions Dietitian-administered counseling was more effective than providing literature in reducing dietary sodium intake in patients with stable HF. (Am Heart J 2005;150:716.e1-716.e5.)

Nonadherence with medical and nutritional therapies is a major and potentially reversible cause of adverse events for patients living with heart failure (HF).¹⁻³ The multidisciplinary HF clinic has become the cornerstone for the management of this syndrome in large part to improve patient adherence and to reduce hospital readmission rates.⁴⁻¹² Dietary modification is a standard therapy in most HF programs, with the primary nutritional prescription being sodium restriction. Various methods of implementing a sodium-restricted diet have been used with nurse-delivered dietary education being most common. Participation of a dietitian in the education of patients with HF has been well described^{7-9,13}; however, only a few trials have used dietitians to deliver nutrition education.^{7-9,13,14} Dietitian involvement is often limited

to patients with nutritionally complex problems such as morbid obesity, diabetes, or severe volume overload. Dietitian counseling may lead to improved adherence to a sodium-restricted diet, and possibly to overall nutritional status. This latter possibility is important based on the recent recognition of malnutrition in this setting.¹⁵

Despite the potential benefits of dietitian involvement in the management of HF, whether counseling administered by a dietitian is more effective in reducing sodium intake than usual care has not been tested in a controlled trial. This is relevant because usual care for patients observed in HF clinics typically already includes educational materials and general nutritional counseling by nurses and/or physicians. Demonstration of the added benefit of dietitian-delivered counseling would also provide justification for the cost of widespread specialized nutritional counseling as a component of a multidisciplinary clinic. Therefore, the purpose of this study was to compare 2 strategies to prescribe a sodium-restricted diet to patients with stable diuretic-requiring HF: counseling by a registered dietitian versus standard delivery of written educational materials to patients.

Methods

Study population

Fifty consecutive patients were recruited from an ambulatory heart function clinic at Mount Sinai Hospital in Toronto, a

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Table 1. Responses to dietitian education versus usual care at 3 months

	Dietitian education		Usual care	
	Baseline	3 Months	Baseline	3 Months
Sodium intake (g/d)	2.80 ± 0.30	2.14 ± 0.23*	3.00 ± 0.31	2.74 ± 0.35
Sodium intake (mg/kJ/d)	0.40 ± 0.04	0.30 ± 0.02*	0.35 ± 0.02	0.36 ± 0.03
Fluid intake (L/d)	1.86 ± 0.11	1.88 ± 0.13	2.26 ± 0.21	2.02 ± 0.15
Fluid intake (mL/kJ/d)	0.27 ± 0.01	0.27 ± 0.02	0.26 ± 0.02	0.26 ± 0.02
Energy intake (kJ/d)	6884 ± 339	7045 ± 537	8512 ± 599	7599 ± 562
Carbohydrate intake (g/d)	220 ± 19	213 ± 21	247 ± 18	226 ± 16
Protein intake (g/d)	79.4 ± 4.1	85.4 ± 9.8	87.8 ± 6.86	86.9 ± 7.27
Fat intake (g/d)	53.7 ± 4.4	59.3 ± 5.3	81.3 ± 7.4	65.7 ± 7.1
Weight (kg)	82.8 ± 3.3	82.7 ± 3.3	82.5 ± 4.1	82.4 ± 4.0
Serum Na (mmol/L)	138 ± 1	137 ± 1	139 ± 1	137 ± 1
BP systolic (mm Hg)	114 ± 4	110 ± 4	114 ± 4	116 ± 4
BP diastolic (mm Hg)	71 ± 2	68 ± 2	71 ± 2	74 ± 3
Heart rate (bpm)	72 ± 3	72 ± 3	74 ± 3	73 ± 3

Data given as mean ± SE.

**P* < .05 compared with baseline by 2-way repeated-measures ANOVA.

tertiary care, academic center. Patients were eligible to participate if they had stable HF with a left ventricular ejection fraction (LVEF) <35%, were taking at least 20 mg furosemide daily, had stable symptoms without hospitalization or change in diuretic dose in the past 3 months, and were able to read and comprehend a nutrition education package. Patients were excluded if their primary residence was an institution where meal and food selections were not by choice, if they had received nutrition counseling from a dietitian in the past 6 months, had diabetes mellitus requiring insulin, significant renal dysfunction (creatinine > 185 mmol/L), or were unable to give informed consent.

Randomization and study design

Patients were randomized into 1 of 2 groups: dietitian education or usual care (control). Both groups were prescribed a 2 g/d sodium-restricted diet and were provided with standardized nutrition education materials. Only patients in the dietitian education group received 2 individualized nutrition-counseling appointments with a registered dietitian. Study end points were collected 3 months after randomization. The study protocol was approved by the Research Ethics Board at Mount Sinai Hospital and all patients provided written informed consent.

Nutrition education materials

The nutrition education package, which was given to all patients, included 2 components: a 2 g/d sodium-restricted diet handout and a low-sodium shopper's guide. These materials were developed by a group of dietitians and were reviewed by a communications specialist to maximize readability. The low-sodium handout was developed specifically for patients with HF and outlined the rationale of salt restriction. It included general information on sodium/salt, discussed high-/low-sodium foods, and provided low-sodium recipes. The handout also gave instruction on label reading, tips for eating in restaurants, and seasoning foods without adding salt. The low-sodium shopper's guide was a booklet that reviewed sodium content in commonly purchased foods. It also listed low-sodium versions of foods and provided a list of local venues that carry these products.

Nutrition education provided to the dietitian education group

The nutrition education protocol was developed by dietitians at Mount Sinai Hospital. Nutrition counseling focused on the 2 g/d sodium-restricted diet prescription. The initial session occurred 1 week after recruitment and after completion of baseline food diaries. At this time, patients received the nutrition education materials (see above). The dietitian completed a nutrition assessment to determine dietary patterns and trends in sodium intake. Patients were educated on the concepts and strategies for a sodium-restricted diet and individualized nutrition care plans and goals were developed. The first session lasted 45 minutes. Follow-up counseling occurred 4 to 6 weeks after the initial visit with the dietitian. At this visit, previous teaching and nutrition goals were reviewed. This visit lasted 30 minutes.

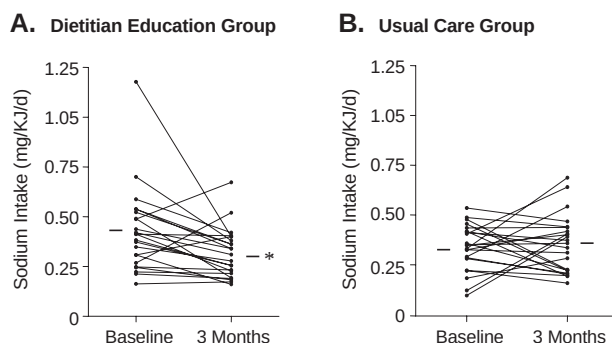
Usual care provided to the control group

On the day of randomization the HF clinic nurse completed a routine assessment, which in our group includes a review of fluid intake and avoidance of high-sodium foods. Patients were told that the educational package contained information and guidelines regarding sodium intake and were asked to open and read the contents at home, after completion of the baseline food diaries. Study personnel informed patients in the usual care group that the educational package described a 2-g sodium-restricted diet. No further counseling was offered before collecting study end points at 3 months.

Study end points

The primary study end point was change in dietary sodium. Secondary nutritional end points included fluid, energy, carbohydrate, protein, and fat intakes. Secondary clinical end points included weight, diuretic dose, and blood pressure. A 3 day food record was used to measure intake and evaluate dietary adherence. This is a common tool used in clinical trials to assess dietary intake and 3 days is sufficient to capture individual day-to-day variation.¹⁶ Patients were instructed how to complete a food record including estimation of portion sizes,

Figure 1



Individual patient dietary sodium daily intake adjusted for total energy intake and mean values (–) at baseline and 3 months. The dietitian education group (A) and the usual care group (B). * $P < .05$ versus baseline by 2-way repeated-measures analysis of variance.

units of measurement, and description of food items. Patients recorded 2 weekdays and 1 weekend day in their diaries. Both study groups received the same instructions for the recording of food diaries. All food records were analyzed using ESHA Food Processor Version 7.4 (Salem, Ore).

Statistical analysis

Baseline comparisons were performed with t tests and χ^2 . Within- and between-group comparisons of the impact of dietitian counseling were performed using a 2-way repeated-measures analysis of variance and the Student-Newman-Keuls post hoc comparison. All data are expressed as mean $\pm$ SE. All statistical analyses were performed using SigmaStat version 1.0 (Systat Software Inc., Richmond, Calif). A P value of $<.05$ was accepted as significant.

Results

Baseline characteristics

The dietitian education group (mean age 56 ± 3 years, 18 men) had a mean LVEF of $22\% \pm 2\%$. This group included 12 patients with coronary disease and 11 patients with non-ischemic cardiomyopathy. The mean furosemide dosage in this group was 82 ± 13 mg/d. The usual care group (mean age 61 ± 3 years, 17 men) had a mean LVEF of $23\% \pm 2\%$. Nine patients in this group had coronary disease, and the mean furosemide dosage was 90 ± 12 mg/d. There were no differences between the groups at baseline in terms of clinical characteristics or dietary profiles (see Table I). The mean daily sodium intakes were ≤ 3 g in both groups (see Table I).

Although 50 patients were recruited, 3 patients were excluded from the final analysis: 2 for not completing food records and 1 patient developed hypothyroidism and resigned from the study. Therefore, the final analysis included 23 patients in the dietitian education group and

24 in the usual care group. No patients required additional clinic visits or visits to the emergency department during this 3-month investigation. We did not record or control patient visits to family physicians.

Responses to dietitian education versus usual care

Patients in the dietitian education group had a significant reduction in dietary sodium at 3 months ($P < .05$), a response that was also significant when dietary sodium was adjusted for total energy intake ($P < .05$) (Table I and Figure 1). This reduction in daily sodium intake occurred without any changes in total fluid intake or the macronutrient content of patients' diets (Table I). As fluid and overall nutritional intake did not change, there was also no change in weight in this group. There was a trend for reduced blood pressure in the intervention group at 3 months that did not achieve statistical significance.

In the usual care group there were no significant changes in dietary sodium intake alone or when adjusted for energy (Table I and Figure 1). There were also no changes in any other nutritional end points. No patient in either group required any adjustment in furosemide or other medical therapy during the course of this study.

Discussion

The central finding of this investigation was that nutrition counseling administered by a registered dietitian resulted in a reduction in dietary sodium intake in patients with stable HF. In contrast, simply providing literature on a sodium-restricted diet was ineffective in reducing sodium intake in this population. This is the first controlled and randomized demonstration of the effectiveness of dietitian-administered counseling, independent of other interventions that occur in the setting of multidisciplinary HF management.

The effectiveness of dietary education provided by a dietitian, as compared with simply providing educational materials, is not surprising based on the complexities associated with adherence to a low-sodium diet. A recent survey of patients with HF highlighted some of the challenges posed by a low-sodium diet.¹⁷ These included lack of knowledge of low-sodium foods, difficulty with adherence to sodium restriction when eating in restaurants or social events, and limited food choices compatible with a low-sodium diet. Furthermore, low-sodium foods tend to be more expensive and less accessible. The expertise of a dietitian allows for detailed nutrition education to deal with these issues. Therefore, the success of our intervention was likely attributed to the dietitians' ability to teach skills and strategies for incorporating a sodium-restricted diet into a patient's lifestyle.

Although not the focus of this study, patients with HF often have comorbidities that can also be addressed with

nutritional therapy. These include renal dysfunction, diabetes mellitus, hyperlipidemia, and morbid obesity. Furthermore, because a reduction in dietary sodium intake necessitates a change in food patterns, secondary alterations in macronutrient and micronutrient intakes may result.^{18,19} This is especially relevant because patients with HF may not be meeting their nutritional requirements.²⁰ Therefore, dietitian-administered counseling may also assist patients to achieve optimal nutritional health while complying with a low-sodium diet.

Sodium restriction is the primary nutritional therapy for the management of HF. A sodium diet of 2 to 3 g/d is recommended based on consensus reports^{21,22}; however, no clinical trials have evaluated this guideline. Therefore, the long-term benefits of sodium restriction on clinical end points such as diuretic requirements and repeat hospitalizations remain to be established. Mechanistic studies in patients with HF have confirmed that sodium restriction reduces extracellular volume,²³ and that patients with HF have impaired sodium homeostasis which is amenable to sodium restriction.²⁴ However, Alvelos et al²⁵ recently demonstrated increases in plasma norepinephrine and aldosterone in euolemic patients with HF in response to a 2.3 g/d sodium diet, raising debate as to the most appropriate sodium prescription in this population. Despite the uncertainty regarding which patients should restrict sodium and to what extent sodium should be reduced, we elected to aim for a 2 g/d sodium diet in this study based on current recommendations.^{21,22} Furthermore, we limited inclusion to diuretic-requiring patients given that reduced sodium intake would likely be most clinically relevant to this group.

In this study we used a 3-day food record to measure sodium intake. Urinary sodium provides an accurate reflection of sodium intake²⁶; however, based on high day-to-day variability in dietary sodium it has been estimated that three to fourteen 24-hour collections would be needed to estimate true daily intake with <10% variation.^{27,28} Food records likely have similar limitations but have the advantage of being easily administered. Food record assessment of sodium intake has been validated against urinary sodium excretion in healthy individuals and in patients with hypertension.^{26,29} The validity of these methods in the setting of HF is untested. Given the variability of sodium content in food, patients were required to record whether food items were commercially or home prepared and to indicate specific brands of foods consumed and names of restaurants attended. This comprehensive approach and the use of validated specialized nutritional analysis software optimized the assessment of dietary sodium and nutritional intake.³⁰ In the present study, although only one group received dietary counseling, both groups received similar instructions from a dietitian regarding the completion of multiple day food records.

Limitations of the present study should be considered. This was a small study that was performed in a single center; therefore, its generalizability cannot be assumed. However, this study included a control group, recruited patients similar to other HF studies, and performed an intervention that was not extraordinary. In this study, there were no changes to hemodynamic, anthropometric, and laboratory end points. Based on the small sample size, the relatively short study period, and the stability of the patients recruited for this study, changes in these parameters were unlikely. Because patients were assessed at 3 months, it is unknown if these results can be sustained over longer periods. In some patients, ongoing counseling may be required. We also did not determine whether the control group actually read the self-help literature provided to them. However, this is reflective of real-life experience.

These results provide evidence for the role of a dietitian as part of the HF multidisciplinary team in helping patients achieve and adhere with a sodium-restricted diet. Although the cost-effectiveness of a dietitian is unknown in this population, improved adherence to nutritional prescriptions in this study provides evidence for the role of dietitians in the management of patients with HF.

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